

1 Bounding the Norm of the Action-Value Function

Given a finite MDP with optimal policy π^* and optimal action-value function $q^*(s, a)$. Let $q(s, a)$ be another state-action value function and define its greedy policy as

$$\pi_q(s) = \arg \max_a q(s, a).$$

Assume the discount factor $0 < \gamma < 1$, and both π^* and π_q are deterministic policies.

Define the infinity norms

$$\|q\|_\infty := \max_{(s,a)} |q(s, a)|, \quad \|v\|_\infty := \max_s |v(s)|.$$

(a)

Suppose the reward satisfies

$$|r| \leq r_{\max}, \quad \forall r \in R.$$

Show that for any policy π ,

$$\|q^\pi\|_\infty \leq \frac{r_{\max}}{1 - \gamma}.$$

$$\begin{aligned} q^\pi(s, a) &= E_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid s, a \right] \\ &\because r \leq r_{\max} \quad \forall r \in R \\ \therefore \|q^\pi\|_\infty &\leq \sum_{t=0}^{\infty} \gamma^t r_{\max} = r_{\max} \frac{1}{1 - \gamma} \\ \therefore \|q^\pi\|_\infty &\leq \frac{r_{\max}}{1 - \gamma} \end{aligned}$$

(b)

Let $v^*(s)$ be the optimal value function and $v^{\pi_q}(s)$ be the value function under policy π_q .

i. Show that for all $s \in S$,

$$v^*(s) - v^{\pi_q}(s) = q^*(s, \pi^*(s)) - q^*(s, \pi_q(s)) + \gamma \sum_{s'} p(s'|s, \pi_q(s)) [v^*(s') - v^{\pi_q}(s')].$$

Prove: because all policies are deterministic.

$$\begin{aligned} v^*(s) - v^{\pi_q}(s) &= q^*(s, \pi^*(s)) - q^{\pi_q}(s, \pi_q(s)) = q^*(s, \pi^*(s)) - q^*(s, \pi_q(s)) + \\ &+ q^*(s, \pi_q(s)) - q^{\pi_q}(s, \pi_q(s)) \end{aligned}$$

$$\begin{aligned}
q^*(s, \pi_q(s)) - q^{\pi_q}(s, \pi_q(s)) &= R + \gamma \sum p(s' | s, \pi_q(s)) v^*(s') - R - \gamma \sum p(s' | s, \pi_q(s)) v^{\pi_q}(s') \\
&= \gamma \sum p(s' | s, \pi_q(s)) (v^*(s') - v^{\pi_q}(s')) \\
\therefore v^*(s) - v^{\pi_q}(s) &= q^*(s, \pi^*(s)) - q^*(s, \pi_q(s)) + \gamma \sum p(s' | s, \pi_q(s)) [v^*(s') - v^{\pi_q}(s')]
\end{aligned}$$

ii. Show that

$$v^*(s) - v^{\pi_q}(s) \leq 2\|q - q^*\|_\infty + \gamma\|v^{\pi_q} - v^*\|_\infty.$$

$$\because \pi_q(s) = \arg \max_a q(s, a), \quad \pi^*(s) = \arg \max_a q^*(s, a).$$

$$\therefore q(s, \pi_q(s)) \geq q(s, \pi^*(s)), \quad q^*(s, \pi^*(s)) \geq q^*(s, \pi_q(s))$$

Then:

$$q^*(s, \pi^*(s)) - q^*(s, \pi_q(s)) = [q^*(s, \pi^*(s)) - q(s, \pi^*(s))] + [q(s, \pi_q(s)) - q^*(s, \pi_q(s))] + [q(s, \pi^*(s)) - q(s, \pi_q(s))]$$

$$\leq [q^*(s, \pi^*(s)) - q(s, \pi^*(s))] + [q(s, \pi_q(s)) - q^*(s, \pi_q(s))] \leq \|q - q^*\|_\infty + \|q - q^*\|_\infty = 2\|q - q^*\|_\infty$$

$$\gamma \sum p(s' | s, \pi_q(s)) [v^*(s') - v^{\pi_q}(s')] \leq \gamma \sum p(s' | s, \pi_q(s)) \|v^{\pi_q} - v^*\|_\infty$$

$$\because \sum p(s' | s, \pi_q(s)) = 1$$

$$\therefore \gamma \sum p(s' | s, \pi_q(s)) [v^*(s') - v^{\pi_q}(s')] \leq \gamma\|v^{\pi_q} - v^*\|_\infty$$

So we can combine two parts:

$$v^*(s) - v^{\pi_q}(s) \leq 2\|q - q^*\|_\infty + \gamma\|v^{\pi_q} - v^*\|_\infty.$$

iii. Show that

$$\|v^* - v^{\pi_q}\|_\infty \leq \frac{2\|q - q^*\|_\infty}{1 - \gamma}.$$

$$\because v^*(s) \geq v^{\pi_q}(s) \quad \forall s$$

$$\therefore v^*(s) - v^{\pi_q}(s) = \|v^* - v^{\pi_q}\|_\infty$$

So:

$$\|v^* - v^{\pi_q}\|_\infty \leq 2\|q - q^*\|_\infty + \gamma\|v^* - v^{\pi_q}\|_\infty$$

$$\|v^* - v^{\pi_q}\|_\infty \leq \frac{2\|q - q^*\|_\infty}{1 - \gamma}$$

In other words, when q is close to the optimal q^* , the value function of its greedy policy π_q is also close to the optimal value function.

(c)

Consider value iteration on the action-value function:

$$q^{(k+1)} = T_{\max}^q q^{(k)}, \quad q^{(0)} = 0,$$

where $T_{\max}^q$ is the Bellman optimality operator and

$$q^* = T_{\max}^q q^*.$$

Show that when

$$k \geq \left(\log \frac{1}{\gamma} \right)^{-1} \log \frac{2r_{\max}}{(1-\gamma)^2 \epsilon},$$

we have

$$\|v^{\pi_{q^{(k)}}} - v^*\|_{\infty} \leq \epsilon.$$

(Hint: use the fact that $T_{\max}^q$ is γ -contracting with respect to $\|\cdot\|_{\infty}$.)

$$\|v^{\pi_{q^{(k)}}} - v^*\|_{\infty} \leq \frac{2\|q^{(k)} - q^*\|_{\infty}}{1-\gamma}$$

Because $T_{\max}^q$ is γ -contracting with respect to $\|\cdot\|_{\infty}$

So:

$$\|q^{(k)} - q^*\|_{\infty} \leq \gamma^k \|q^{(0)} - q^*\|_{\infty} = \gamma^k \|q^*\|_{\infty}$$

From (a), we get: $\|q^*\|_{\infty} \leq \frac{r_{\max}}{1-\gamma}$

$$\text{Therefore: } \|v^{\pi_{q^{(k)}}} - v^*\|_{\infty} \leq \frac{2\gamma^k}{1-\gamma} \frac{r_{\max}}{1-\gamma} = \frac{2r_{\max}}{(1-\gamma)^2} \gamma^k$$

Let it $\leq \epsilon$

$$\gamma^k \leq \frac{\epsilon(1-\gamma)^2}{2r_{\max}}$$

$$k \cdot \log \gamma \leq \log \frac{\epsilon(1-\gamma)^2}{2r_{\max}} \because \gamma \in (0, 1), \therefore \log \gamma < 0$$

$$\therefore k \geq \log \frac{\epsilon(1-\gamma)^2}{2r_{\max}} / \log \gamma = \log \frac{2r_{\max}}{\epsilon(1-\gamma)^2} / \log \frac{1}{\gamma} = (\log \frac{1}{\gamma})^{-1} \log \frac{2r_{\max}}{\epsilon(1-\gamma)^2}$$

So it is proved that when $k \geq (\log \frac{1}{\gamma})^{-1} \log \frac{2r_{\max}}{\epsilon(1-\gamma)^2}$

$$\|v^{\pi_{q^{(k)}}} - v^*\|_{\infty} \leq \epsilon$$

2 Error Propagation in Approximate MDP Models

Consider an MDP

$$M = (S, A, R, p, \gamma)$$

with discount factor $0 < \gamma < 1$.

Suppose we construct an approximate MDP

$$\hat{M} = (S, A, R, \hat{p}, \gamma).$$

The transition probabilities satisfy

$$\hat{p}(s'|s, a) = p(s'|s, a), \quad \forall s', s, a.$$

The reward functions satisfy

$$|\hat{r}(s, a) - r(s, a)| \leq \epsilon,$$

equivalently

$$\left| \sum_r r \hat{p}(r|s, a) - \sum_r r p(r|s, a) \right| \leq \epsilon, \quad \forall s, a.$$

(a)

Let $v^*(s)$ and $\hat{v}^*(s)$ be the optimal value functions of M and $\hat{M}$, respectively.

Show that

$$\|v^* - \hat{v}^*\|_\infty \leq \frac{\epsilon}{1 - \gamma}.$$

(Hint: let $T_{\max}$ and $\hat{T}_{\max}$ be the Bellman optimality operators of M and $\hat{M}$. Then $v^* = T_{\max} v^*$ and $\hat{v}^* = \hat{T}_{\max} \hat{v}^*$.)

$$\begin{aligned} v^* &= T_{\max} v^*, \quad \hat{v}^* = \hat{T}_{\max} \hat{v}^* \\ \|v^* - \hat{v}^*\|_\infty &= T_{\max} v^* - \hat{T}_{\max} \hat{v}^* = T_{\max} v^* - T_{\max} \hat{v}^* + T_{\max} \hat{v}^* - \hat{T}_{\max} \hat{v}^* \\ \text{According to the } \gamma\text{-contraction of } T_{\max} \\ T_{\max} v^* - T_{\max} \hat{v}^* &= \gamma \|v^* - \hat{v}^*\|_\infty \end{aligned}$$

$$\begin{aligned} T_{\max} \hat{v}^* - \hat{T}_{\max} \hat{v}^* &= r(s, a) + \gamma \sum p(s' | s, a) \hat{v}(s') - \hat{r}(s, a) - \gamma \sum \hat{p}(s' | s, a) \hat{v}(s') = \\ r(s, a) - \hat{r}(s, a) &\leq \epsilon \end{aligned}$$

$$\begin{aligned} \therefore \|v^* - \hat{v}^*\|_\infty &\leq \gamma \|v^* - \hat{v}^*\|_\infty + \epsilon \\ \text{So } \|v^* - \hat{v}^*\|_\infty &\leq \frac{\epsilon}{1 - \gamma} \end{aligned}$$

(b)

Suppose

$$\hat{r}(s, a) = r(s, a) + \epsilon, \quad \forall s, a.$$

Show that

$$\hat{v}^*(s) = v^*(s) + \frac{\epsilon}{1-\gamma}, \quad \forall s.$$

$$\begin{aligned} \hat{v}^*(s) &= E[\sum_{t=0}^{\infty} \gamma^t \hat{r}_t(s, a) \mid s_0 = s] \\ &= E[\sum_{t=0}^{\infty} \gamma^t r_t(s, a) + \epsilon \mid s_0 = s] \\ &= E[\sum_{t=0}^{\infty} \gamma^t r_t(s, a) \mid s_0 = s] + \sum_{t=0}^{\infty} \gamma^t \epsilon \\ &= v^*(s) + \frac{\epsilon}{1-\gamma} \\ \therefore \hat{v}^*(s) &= v^*(s) + \frac{\epsilon}{1-\gamma} \end{aligned}$$